
BEYOND SINGLE DEFENSES: COMBINING ADVERSARIAL TRAINING AND RANDOMIZED SMOOTHING DEFENSES *

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1 Introduction

In machine learning, deep neural networks are highly vulnerable to adversarial examples - inputs modified by adding small, imperceptible perturbations that can fool a model in to producing an incorrect prediction. A canonical example is adding carefully computed noise to an image of a stop sign, causing an autonomous vehicle's perception system to misidentify it as a yield sign - a failure with catastrophic real-world consequences. As neural networks increasingly deployed in real-world and safety-critical settings, building robust models that can resist such attacks has become a central challenge in adversarial machine learning.

In this work, we investigate a simpler and more deployment-realistic alternative: applying adversarial training and randomized smoothing sequentially rather than jointly. Specifically, we first train a base classifier using Projected Gradient Descent (PGD) adversarial training, then apply randomized smoothing at inference time without any modification to the training procedure. We evaluate this pipeline on MNIST and CIFAR-10 datasets, measuring clean accuracy, empirical robust accuracy under ℓ_2 -norm PGD attacks, and certified accuracy at $r = 0.5$, comparing a standard-trained baseline and a Gaussian augmentation baseline.

2 Prior Work

Adversarial robustness in neural networks has split into two complementary paradigms: empirical and certified defenses. Empirical defenses seek to improve robustness against by training against a set of known-attacks while certified defenses provide provable guarantees that no perturbation within a bounded region can alter a model's prediction. Our work is at the intersection of both.

Adversarial training (AT) is the most popular defense technique in empirical defenses. Madry et al. (2018) formalized the problem as a min-max robust optimization objective - minimizing loss over the worst-case perturbation (with maximized sample loss) found by Projected Gradient Descent (PGD). Training on PGD-generated adversarial examples ensures that the resulting model is empirically robust against first-order adversarial attacks. However, adversarial training only yields empirical robustness and provides no provable guarantees against unseen attacks. Moreover, adversarially-trained models tend to sacrifice clean accuracy for robustness.

Randomized Smoothing (RS) - introduced by Cohen et al. (2019) - provides a scalable certified defense against ℓ_2 norm-bound perturbations. Given a base classifier f and noise level $\sigma > 0$, the smoothed classifier g is defined as

$$g(x) = \arg \max_{c \in \mathcal{Y}} \Pr_{\eta \sim \mathcal{N}(0, \sigma^2 I)} [f(x + \eta) = c].$$

Cohen et al. proved that if the top-class probability p_A and the runner-up probability p_B satisfy $p_A > p_B$, then g is certifiably robust within the ball of ℓ_2 radius.

$$R = \frac{\sigma}{2} (\Phi^{-1}(p_A) - \Phi^{-1}(p_B)),$$

where Φ^{-1} is the inverse standard normal CDF. In practice, since p_A and p_B are unknown, they are estimated with Monte Carlo sampling and one-sided confidence bounds.

Salman et al. (2019) demonstrated that adversarially trained base classifiers significantly improve certified accuracy under randomized smoothing. Rather than applying standard PGD, they introduced **SmoothAdv** - a specialized attack

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that directly targets the smoothed classifier by maximizing loss over the expectation of Gaussian-corrupted inputs. Training against this stronger, smoothing-aware attack yielded state-of-the-art certified accuracy on CIFAR-10.

In contrast to SmoothAdv and related methods that couple AT and RS into a single joint training objective, our work applies them sequentially - first training with PGD adversarial training and then applying randomized smoothing at inference time without modifying the base classifier. This decoupled design reflects a more-practical deployment scenario where retraining from scratch with a custom procedure is not always feasible.

3 Methodology

3.1 Model

For our experiments, our model was a *SmallCNN* model - a lightweight convolutional network consisting of four convolutional layers with ReLU activations, followed by a two-layer fully connected head that maps to class logits. In two of the convolutional layers, spatial downsampling is performed via strided convolutions instead of pooling. The architecture is input-agnostic, accommodating both MNIST ($1 \times 28 \times 28$) and CIFAR-10 ($3 \times 32 \times 32$) inputs. All different configurations of model defenses and attacks share the same model for their base classifier.

3.2 Training Conditions

We evaluated four different training procedures:

- **Standard:** Regular model training with no defenses.
- **Adversarial Training:** Model is trained using PGD adversarial training - a 20-step PGD inner loop with $\epsilon \in \{0.1, 0.5, 1.0\}$ and $\alpha = \epsilon/8$. No randomized smoothing is applied.
- **Randomized Smoothing:** Model is trained with Gaussian noise injection ($\sigma = 0.25$) at training time, following the baseline training procedure of Cohen et al. (2019). Randomized smoothing certification is applied at inference.
- **Adversarial Training + Randomized Smoothing (ours):** The model is trained with adversarial training with the same parameters as above. Afterwards, the model is also trained with Gaussian noise injection of $\sigma = 0.25$. Randomized smoothing certification is applied at inference time. This is our variant - a sequential composition of both defenses.

All models were trained with the following hyper-parameters: 20 epochs, batch size of 128, learning rate of 1×10^{-3} . All of our training runs and evaluation had a fixed seed of 42, enabling consistency and reproducibility. The optimizer for training was the Adaptive Moment Estimation (Adam) optimizer. All of training was done on a Macbook Pro with Torch MPS.

3.3 Evaluation

Each training condition is evaluated across three metrics:

- **Clean Accuracy:** Standard accuracy on unperturbed images.
- **Empirical Robust Accuracy:** Accuracy under PGD- ℓ_2 attack ($\epsilon \in \{0.1, 0.5, 1.0\}$, steps = 20, $\alpha = \epsilon/8$). If a model is *RS*-trained, we use the Expectation over Transformation (EOT) variant. Rather than computing gradients through a single forward pass, EOT averages gradients over multiple Gaussian-corrupted samples to directly attack the smoothed classifier g

$$\nabla_x \mathbb{E}_{\eta \sim \mathcal{N}(0, \sigma^2 I)} [\mathcal{L}(f(x + \delta + \eta), y)]$$

This makes EOT more appropriate for stochastic defenses like randomized smoothing, as it accounts for the Gaussian noise distribution rather than treating the defense as deterministic.

- **Certified Accuracy @ r :** Fraction of test inputs correctly classified with a certified ℓ_2 radius $\geq r$, computed via Cohen et al.’s certification procedure.

3.4 Randomized Smoothing Certification Parameters

For all randomized smoothing models, certification uses the following fixed hyperparameters in the table below. If the top-class probability p_A cannot be lower-bounded above 0.5 at confidence $1-\alpha$ via Clopper-Pearson bounds, the classifier abstains rather than returning an unreliable certificate. Abstention rate is reported along certified accuracy.

Parameter	Value
σ (noise level)	0.25
N_0 (selection samples)	100
N (certification samples)	1,000
α (confidence level)	0.001
R (certified radius threshold)	0.5
Number of test images certified	500

Table 1: Randomized Smoothing Certification Hyperparameters

3.5 Datasets

For evaluation, we tested our models on two standard benchmark datasets for image classification: MNIST and CIFAR-10.

The MNIST (Modified National Institute of Standards and Technology) dataset consists of 70,000 grayscale images of handwritten digits (0–9), each of size 28×28 pixels. The dataset is split into 60,000 training images and 10,000 test images. Each image is associated with one of 10 class labels corresponding to the digits 0 through 9. MNIST is a widely used benchmark for evaluating image classification algorithms due to its simplicity and well-understood structure.

The CIFAR-10 dataset consists of 60,000 color images across 10 object classes (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck), each of size $32 \times 32 \times 3$ pixels (RGB). The standard split allocates 50,000 images for training and 10,000 for testing, with exactly 6,000 images per class. CIFAR-10 presents a significantly greater classification challenge due to its higher visual complexity and class variation compared to MNIST.

4 Results and Interpretation

For results, we evaluate four defenses - standard training, adversarial training (AT), randomized smoothing (RS), and their combination (AT+RS) - on MNIST and CIFAR-10, reporting clean accuracy, empirical accuracy under PGD- ℓ_2 and EOT-PGD attacks, and certified accuracy + abstention rates for smoothed models. We'll only discuss results for the ε value in each dataset that produced the most informative behavior. (more graphs in appendix if space allows)

4.1 MNIST

On MNIST, all defenses maintain almost perfect clean accuracy - with standard and defensively-trained models achieving $\approx 99\%$ test accuracy. Under a PGD- ℓ_2 attack, adversarial training dramatically improved robustness by increasing adversarial accuracy from 37% to 95%. Randomized smoothing and hybrid AT-RS models achieve lower empirical robust accuracy against both PGD- ℓ_2 and EOT PGD- ℓ_2 attacks - but still retain high clean accuracy.

For smoothed models on MNIST, smoothed accuracy remains practically constant across clean and adversarial evaluations. Certified accuracy and abstention rates show similar behavior. This limited variation is largely due to certification protocol: for computational reasons, randomized smoothing metrics are estimated on a fixed subset of 500 test images (out of 10,000). Due to this limitation, reported smoothed and certified accuracies should be interpreted as representative of this subset rather than estimates for the full test distribution.

Examining the per-class confidence heatmap, defensively-trained models retain high confidence across each of the digit classes. However, a few classes - such as digits 8 and 9 - noticeably degrade more in performance when under attack. This suggests that robustness is not uniform across the dataset - randomized smoothing and adversarial training mainly stabilize already easy digits.

Model / Evaluation	Accuracy	Smoothed Acc.	Certified Acc.	Abstention Rate
Standard/Clean	0.9900	—	—	—
Standard/ ℓ_2 PGD	0.3702	—	—	—
AT/Clean	0.9917	—	—	—
AT/ ℓ_2 PGD	0.9544	—	—	—
RS/Clean	0.9900	0.990	0.958	0.002
RS/ ℓ_2 PGD	0.9228	0.990	0.958	0.004
RS/ ℓ_2 EOT-PGD	0.9139	0.990	0.958	0.004
AT+RS/Clean	0.9899	0.986	0.950	0.004
AT+RS/ ℓ_2 PGD	0.9243	0.986	0.948	0.002
AT+RS/ ℓ_2 EOT-PGD	0.9054	0.986	0.950	0.002

Table 2: MNIST empirical accuracy, smoothed accuracy, abstention rate, and certified accuracy for $\varepsilon = 1.0$ budget

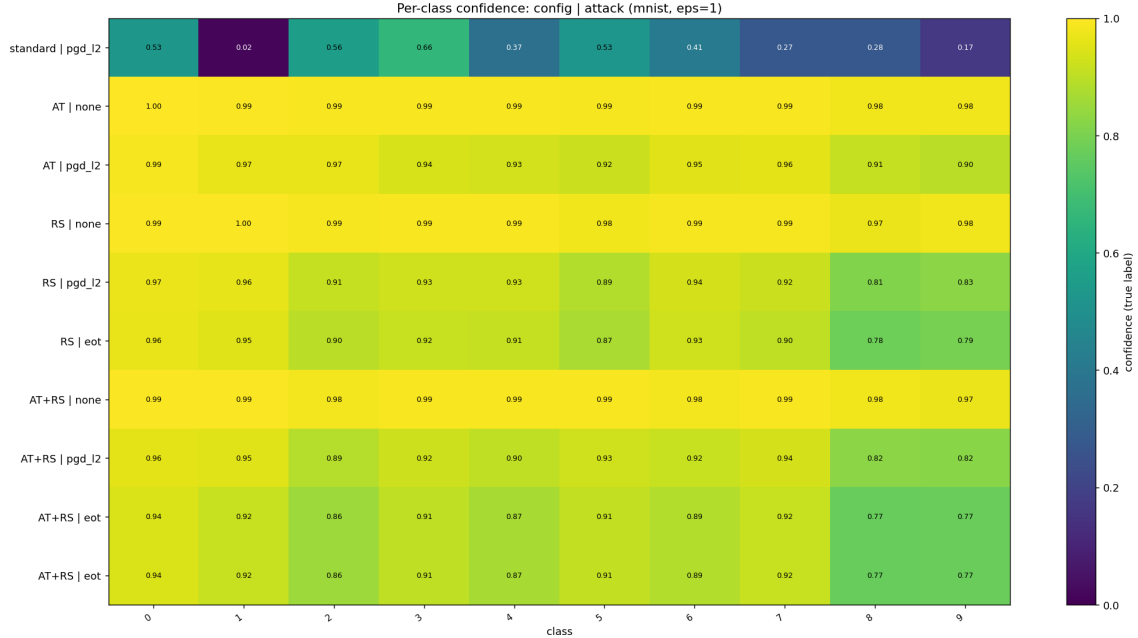


Figure 1: Per-class confidence of MNIST classes for $\varepsilon = 1.0$ (zoom in to see numeric values)

4.2 CIFAR-10

On CIFAR-10, there is much higher variation in performance when comparing different defensively-trained models. Both standard and adversarially-trained models achieved moderate clean accuracy (66% and 57% respectively). But performance degrades heavily under PGD- ℓ_2 attacks, with the standard model dropping $\sim 60\%$ while the AT-model degrades $\sim 15\%$. Randomized smoothing and the hybrid AT-RS models only yield slightly higher accuracies under the same attack. However, EOT attacks degrade empirical accuracy in randomized-smoothing (and hybrid) models more than standard PGD- ℓ_2 attacks. This is to be expected as EOT averages out gradients over multiple noise samples rather than a single noisy sample - optimizing for expected loss and producing a more accurate gradient of the smoothed classifier.

Class categories in CIFAR-10 demonstrate different confidence-levels across various attacks and training configurations. Generally, "animal" classes ("bird", "cat", "deer") show consistently lower confidence when compared to "vehicle" classes ("airplane", "automobile", "ship"). The only exception is the "horse" class, which retains higher accuracy similar to vehicle classes. This suggests that robustness is not only aligned with semantics. Instead, a class like horse may have a more rigid, distinctive shape, or clearer background that behaves like vehicles in feature space. Hence, the class benefits more from adversarial training and randomized smoothing versus other animal classes.

Model / Evaluation	Accuracy	Smoothed Acc.	Certified Acc.	Abstention Rate
Standard/Clean	0.6610	—	—	—
Standard/ ℓ_2 PGD	0.0629	—	—	—
AT/Clean	0.5680	—	—	—
AT/ ℓ_2 PGD	0.4164	—	—	—
RS/Clean	0.6134	0.550	0.198	0.144
RS/ ℓ_2 PGD	0.4224	0.552	0.200	0.140
RS/ ℓ_2 EOT-PGD	0.3544	0.548	0.202	0.142
AT+RS/Clean	0.6029	0.542	0.196	0.166
AT+RS/ ℓ_2 PGD	0.4366	0.544	0.202	0.162
AT+RS/ ℓ_2 EOT-PGD	0.3552	0.544	0.200	0.158

Table 3: CIFAR-10 empirical accuracy, smoothed accuracy, abstention rate, and certified accuracy for $\varepsilon = 0.5$ budget



Figure 2: Per-class confidence of CIFAR-10 classes ($\epsilon = 0.5$) (zoom in to see numeric values)

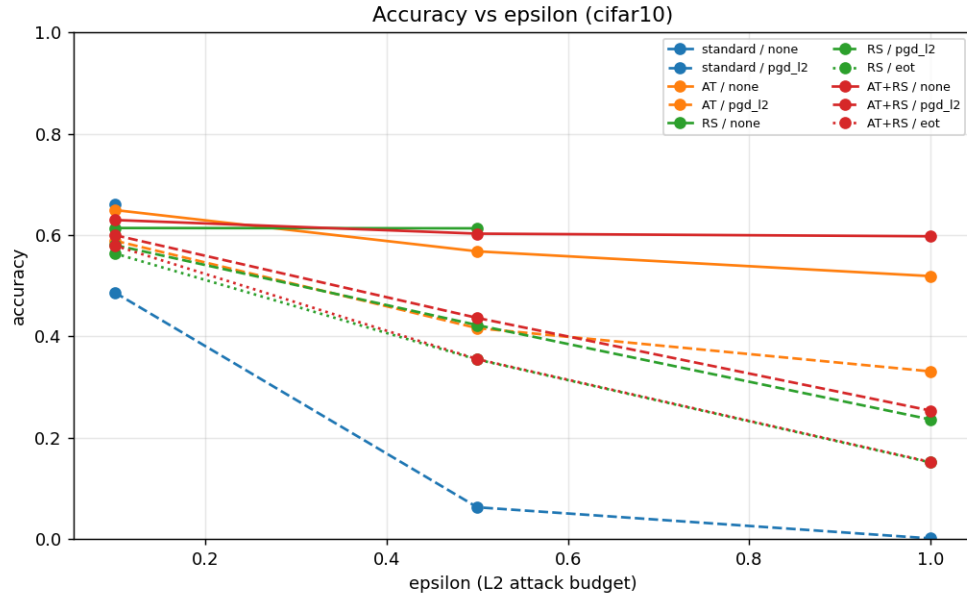


Figure 3: Accuracy vs. epsilon-budget line graph across training/evaluation configurations

5 Limitations and Future Work

The study has several strengths. First, our experimental design was controlled, where we fixed the architecture, attacks, and epsilon budgets to allow fair comparisons between defenses. Recording both robust accuracies and certified/smoothing accuracies enables analysis into trade-off between empirical robustness and provable guarantees. Additionally, using a PGD-EOT attack avoids the pitfall of evaluate smoothed classifiers with only non-adaptive attacks.

At the same time, there are many limitations with our experiments. Due to deterministic training and evaluation (fixed seeding), the randomized smoothing certification only targets the same, small subset of 500 test images and uses identical Monte Carlo sampling patterns for RS and AT+RS. As a result, smoothed accuracy, certified accuracy, and abstention rates show little variation between different training and attack configurations. One might observe higher variability in certification statistics if they were to increase sample a larger number of test samples or randomly resample test subsets altogether.

Additionally, since our model was a small CNN, we limited our performance on the CIFAR-10 dataset. Using a larger model, such as ResNet, may exhibit larger variation in robustness and performance across different attacks and defenses.

Future work should address these limitations. They include the following (1) using a larger model such as ResNet on CIFAR-10 (2) increase Monte Carlo sample counts for tighter certification bounds and lower abstention rate (3) evaluate across multiple random seeds for more robust results and to capture variability across different seeds (4) extend evaluation to additional datasets and a broader range of ℓ_2 radii to obtain more complete robustness and certified accuracy-curves.

6 Conclusion

In conclusion, this work shows that simply composing adversarial training and randomized smoothing in a sequential way does not yield gains in robustness. On CIFAR-10 and MNIST, both empirical and certified evaluations indicate that adversarial training and randomized smoothing each offer meaningful but limited protection. Adversarial training is strongest against standard PGD, while randomized smoothing offers the best certified guarantees and slightly better resilience under EOT-PGD. Their combination largely inherits the weaknesses of both without clearly surpassing either one. Per-class confidence analysis further reveals that robustness is highly uneven across classes, with several animal categories remaining substantially more fragile than vehicle-like classes even under the strongest defenses. Taken together, these findings suggest that achieving useful, uniformly distributed robustness - especially under adaptive attacks - will require more integrated defensive designs and stronger architectures than the straightforward AT+RS pipeline studied here.

References

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- [2] J. Cohen, E. Rosenfeld, and Z. Kolter, “Certified adversarial robustness via randomized smoothing,” in *International Conference on Machine Learning (ICML)*, 2019.
- [3] H. Salman, G. Yang, J. Li, P. Zhang, H. Zhang, I. Razenshteyn, and S. Bubeck, “Provably robust deep learning via adversarially trained smoothed classifiers,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

7 Code

All code for this project is in this Github repository

8 Appendix

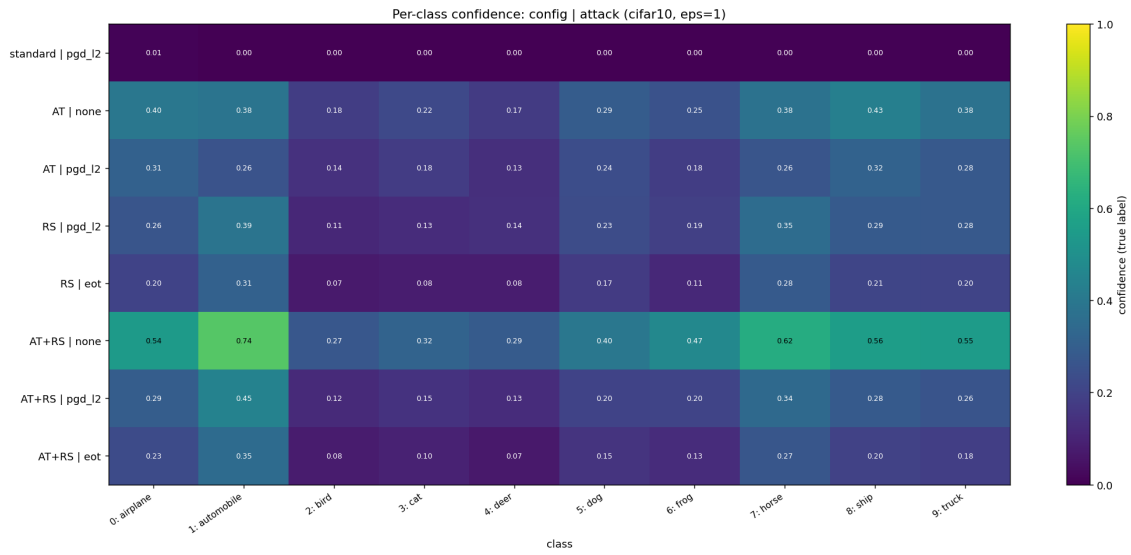


Figure 4: Per-class confidence of CIFAR-10 classes ($\epsilon = 1.0$) (zoom in to see numeric values)

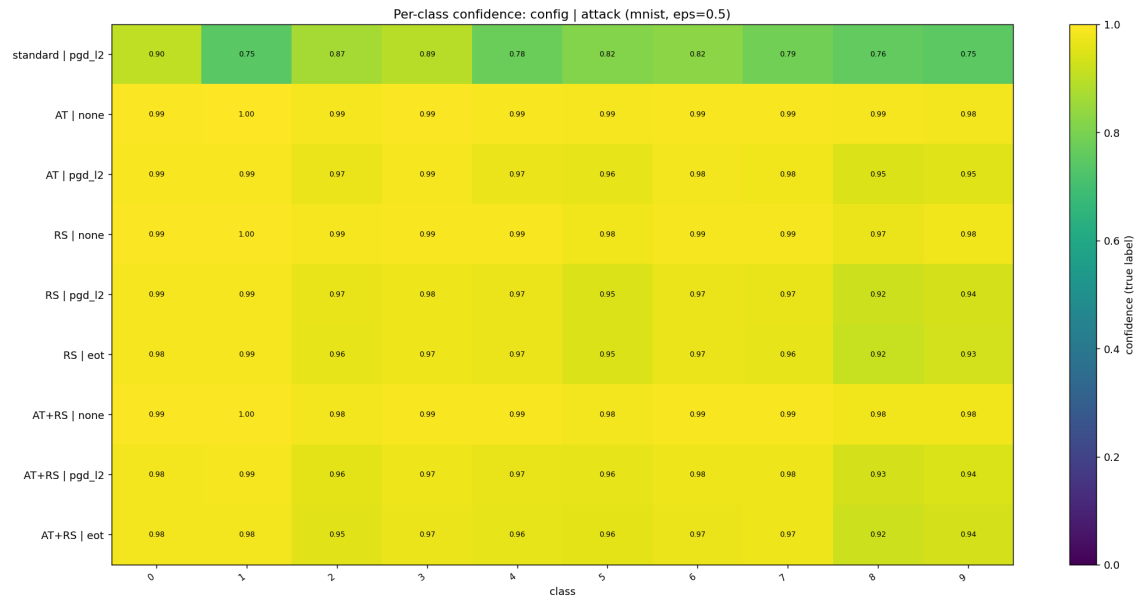


Figure 5: Per-class confidence of MNIST classes ($\epsilon = 0.5$) (zoom in to see numeric values)

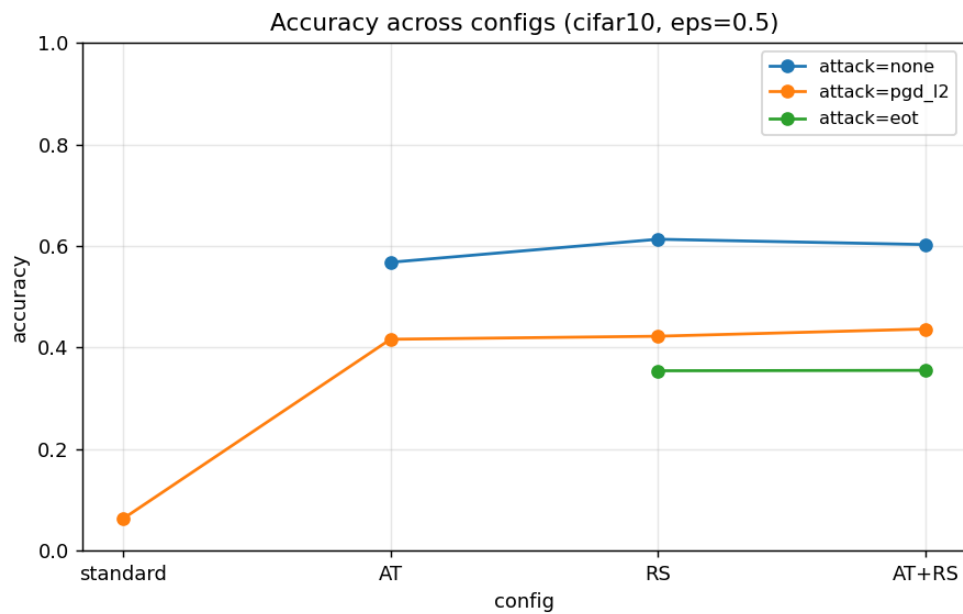


Figure 6: Accuracy vs. Configuration on CIFAR-10, $\epsilon = 0.5$